

# Where Does Being Early Pay?

## A catalogue of testable hypotheses for Paper B

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Each hypothesis below says where the penalty for leading language should be *smaller* (or reversed into an advantage) and where it should be *larger*. The outcome throughout is the one the paper already uses — failure at the five-year proof of continued use, plus the capital-markets and acquisition rungs where noted — and every test is the same comparison: the most leading fifth against the most lagging fifth, within Nice class and year, estimated separately inside the groups the hypothesis names and compared across them. “Ready” means the data exist in the repository; “build” means a new variable must be constructed from data we hold; “new data” means a source we do not yet have.

### Where each literature looked, and who answers whom

| Work                                    | Setting                                                                                                                  | Cites, among these works    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Robinson & Fornell (1985)               | Surveyed consumer-goods businesses (market share by order of entry)                                                      | no reference list deposited |
| Teece (1986)                            | Named innovations and their imitators (theory with cases)                                                                | no reference list deposited |
| Lieberman & Montgomery (1988)           | Catalogue of mechanisms across industries                                                                                | Robinson & Fornell; Teece   |
| Golder & Tellis (1993)                  | Histories of fifty product categories, including diapers, video recorders, personal computers, light beer, safety razors | no reference list deposited |
| Klepper & Miller (1995); Klepper (1997) | New manufactured products through their shakeouts                                                                        | no reference list deposited |
| Lieberman & Montgomery (1998)           | Retrospective; entry timing as a question of resources                                                                   | no reference list deposited |

| Work                       | Setting                                                                                       | Cites, among these works                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Kim & Mauborgne (2004)     | Cases of firms that opened new categories (wine, fitness, circus, air travel, hotels, coffee) | cited by neither of the later works with deposited lists                         |
| Suárez & Lanzolla (2007)   | Reconciliation by pace of technology and market change                                        | Robinson & Fornell; Lieberman & Montgomery 1988 and 1998; Golder & Tellis; Teece |
| Semadeni & Anderson (2010) | Consulting firms' service-mark filings (imitation of new offerings)                           | Lieberman & Montgomery 1988 and 1998; Suárez & Lanzolla                          |

The pioneer-advantage literature answers itself in sequence; the blue-ocean literature runs beside it. Links come from the reference lists publishers deposit with Crossref, so a blank means the list is not deposited, not that the citation is absent. The settings suggest where each claim should hold if it holds anywhere: consumer packaged goods for the pioneer claim, the named histories and shakeout-prone manufactured goods for the counter-claim, and the blue-ocean exemplars' own verticals (hypotheses 17–21).

## A. First-mover mechanisms (Lieberman & Montgomery, 1988)

|   | Claim                                                                                              | Test in the record                                                                                                                                                                   | Data  |
|---|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1 | <i>Technological leadership.</i> Early entrants protected by patents and learning keep their lead. | Lead penalty for owners who also patent (PatentsView match) against those who do not; expect smaller for patenters.                                                                  | ready |
| 2 | <i>Preemption of scarce assets.</i> Early entrants take the best locations and inputs.             | Themes built on physical sites (stores, restaurants, hotels, clinics; classes 35, 43, 44 location vocabulary) against footloose services; expect smaller penalty where sites matter. | build |
| 3 | <i>Buyer switching costs.</i> Customers locked in by contracts stay with the pioneer.              | Descriptions naming subscriptions, memberships, accounts or contracts against one-off goods; expect smaller penalty.                                                                 | build |
| 4 | <i>Network effects.</i> An installed base compounds.                                               | Platform and marketplace vocabulary (platform, marketplace, social network, exchange, community); expect smaller penalty, possibly a reward.                                         | build |

|   | Claim                                                                                        | Test in the record                                                                                                                                                        | Data  |
|---|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5 | <i>Free-riding.</i> Followers copy cheaply what pioneers proved.                             | Themes with low capital intensity and easy imitation (apparel, simple consumer goods, personal services) against capital-heavy manufactured goods; expect larger penalty. | build |
| 6 | <i>Resolution of uncertainty.</i> Pioneers bet before the market settles.                    | Themes whose later share was volatile (surged then receded) against themes that grew steadily; expect larger penalty in volatile themes.                                  | ready |
| 7 | <i>Incumbent inertia.</i> Established firms are slow to follow, so the pioneer's lead lasts. | Lead penalty for debut owners against owners with large existing portfolios filing the same leading language; expect the penalty to fall on new firms.                    | ready |

## B. Pace of change (Suárez & Lanzolla, 2007)

|    | Claim                                                                                                     | Test in the record                                                                                                   | Data  |
|----|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------|
| 8  | <i>Calm waters</i> (slow technology, slow market): a first-mover advantage is durable.                    | Classes and eras with low turnover in the theme mix and flat filing volume; expect the smallest penalty or a reward. | build |
| 9  | <i>Technology leads</i> (fast technology, slow market): early entrants are overtaken by later technology. | Fast theme turnover with flat volume; expect the largest penalty.                                                    | build |
| 10 | <i>Market leads</i> (slow technology, fast market growth): early entrants capture the growing demand.     | Surging filing volume with a stable theme mix (consumer categories, beverages); expect a smaller penalty.            | build |
| 11 | <i>Rough waters</i> (both fast): advantage only for resource-rich entrants.                               | Both fast (technology classes 1995–2000); expect a large penalty, smaller for large or funded owners.                | build |

Pace of technology: year-to-year divergence in a class's theme mix. Pace of market: growth in the class's (or theme's) filing volume. The two cut the class-years into the four quadrants.

## C. Appropriability (Teece, 1986)

|    | Claim                                                                                                                | Test in the record                                                                                                                                                         | Data  |
|----|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 12 | <i>Tight appropriability</i> : where innovations can be protected, the innovator profits.                            | Pharmaceuticals and medical devices (classes 5, 10) and patenting owners; expect smaller penalty.                                                                          | ready |
| 13 | <i>Weak appropriability, generic complementary assets</i> : imitators win.                                           | Services and software with no patenting; expect larger penalty.                                                                                                            | ready |
| 14 | <i>Specialized complementary assets held by incumbents</i> : incumbents who own distribution capture the innovation. | In distribution-heavy consumer goods (food, beverages, cosmetics), the interaction of lead with owner size: leading small entrants fail, leading large incumbents survive. | ready |

#### D. Pioneer disadvantage (Golder & Tellis, 1993) and shakeouts (Klepper, 1997; Klepper & Miller, 1995)

|    | Claim                                                                                                                       | Test in the record                                                                                                                                                          | Data  |
|----|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 15 | <i>Early leaders beat pioneers</i> : the firms remembered as pioneers usually entered a few years after the first entrants. | Within each theme's arrival in a class, the first year's filers against those two to five years in; expect the later cohort to survive better.                              | ready |
| 16 | <i>Survivor bias</i> : pioneer advantage appears only among survivors.                                                      | The lead gradient among all applications, among registrations, and among marks that passed the proof; expect the gradient to fade as the sample is conditioned on survival. | ready |
| 17 | <i>Pioneer advantage in consumer packaged goods</i> (the finding Golder and Tellis dispute).                                | Consumer packaged goods classes (3, 29, 30, 32, 33) against industrial goods; the pioneer literature predicts opposite signs.                                               | ready |
| 18 | <i>Early entrants win shakeouts</i> in manufactured products with scale economies.                                          | Manufactured goods classes (7, 9, 12) against services; and, within surging themes, early joiners against late joiners (already: no order effect).                          | ready |
| 19 | <i>The cited cases</i> : disposable diapers, video recorders, personal computers, light beer, safety razors.                | Lead penalty within the verticals those cases come from (paper and sanitary goods, consumer electronics, beer, razors and blades).                                          | ready |

#### E. Blue ocean (Kim & Mauborgne, 2004)

|    | Claim                                                                                                                                                                               | Test in the record                                                                                                                                                            | Data  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 20 | <i>Reconstructing market boundaries</i> : value comes from combining attributes from different industries.                                                                          | Filings carrying a theme established in another class into their own (new-to-class), and filings whose leading language is a new pairing of existing themes; expect a reward. | ready |
| 21 | <i>The exemplars’ verticals</i> : wine ([yellow tail]), fitness (Curves), live entertainment (Cirque du Soleil), air travel (Southwest, NetJets), hotels, coffee, home improvement. | Lead penalty within each vertical’s class and themes; and where the exemplar’s own filings fall in the lead distribution when they are in the scored years.                   | ready |
| 22 | <i>Creating new demand</i> (non-customers) rather than taking share.                                                                                                                | Themes whose growth came from first-time filers rather than established owners; expect a smaller penalty for the early entrants of such themes.                               | build |

## F. Kinds of new

|    | Claim                                                                                                                                     | Test in the record                                                                                                                                                      | Data  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 23 | <i>New inventions</i> reward pioneers who define the category.                                                                            | Filings carrying a theme new to the whole corpus (too few at 500 themes; use curated invention vocabularies: 3D printing, drones, gene therapy, electric vehicles).     | build |
| 24 | <i>Fads</i> punish late entry, and possibly early entry too.                                                                              | Themes that surged and then fell back below their pre-surge share within five years, against surges that held.                                                          | ready |
| 25 | <i>General-purpose technologies</i> (internet, mobile, cloud, AI) spread across many industries; early adopters in each industry may win. | Vocabularies appearing in ten or more classes; the lead penalty inside each adopting class, by years since the vocabulary reached it (the internet result generalized). | ready |
| 26 | <i>New to the class versus new to the world</i> . Importing an established theme is safer than originating one.                           | New-to-class against new-to-corpus themes, and both against established themes.                                                                                         | ready |
| 27 | <i>Groups of emergent consumer themes</i> (plant-based milk, kombucha, cold brew, hard seltzer, CBD, energy drinks).                      | Pooled into one “emergent consumer category” group, with each category’s early entrants compared with its later entrants and with the class.                            | ready |

## G. Who and where

|    | Claim                                                                                                                                                       | Test in the record                                                                                                                                                  | Data        |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 28 | <i>Clusters speed imitation.</i> Where a theme’s early filers are geographically concentrated (a Silicon Valley or Silicon Alley), followers learn fastest. | Co-location index of the theme’s filers in a five-year window, net of its class (see the clustering measure below); expect a larger penalty in concentrated themes. | ready (ZIP) |
| 29 | <i>Distance from the cluster.</i> Early filers far from a theme’s hubs are shielded from local imitation, or cut off from their resources.                  | Within themes that have a hub, filers in a hub against filers outside; the sign is the question.                                                                    | ready (ZIP) |
| 30 | <i>Foreign pioneers.</i> Foreign firms bringing a theme into the US market face local imitators.                                                            | Foreign-domiciled and Madrid filers against US filers.                                                                                                              | ready       |
| 31 | <i>Resources absorb the penalty.</i>                                                                                                                        | Counsel-represented and large-portfolio owners against self-filed debut owners.                                                                                     | ready       |
| 32 | <i>Speculation.</i> Intent-to-use filings claim a market before entering it.                                                                                | Intent-to-use against use-based filings; expect a larger penalty for speculative leading filings.                                                                   | ready       |
| 33 | <i>Regulation as a barrier.</i> Approval requirements protect the first entrant.                                                                            | Regulated classes (pharmaceuticals, medical devices, alcohol, tobacco, financial services) against unregulated ones.                                                | ready       |
| 34 | <i>Timing of entry in the cycle.</i> Early entry in a boom is punished; in a bust it is rewarded.                                                           | The era swing, extended by the reversal decomposition to every class and theme.                                                                                     | ready       |

## Notes on design

Many of these groupings overlap, and several are proxies (a vocabulary for “switching costs” is not a measured switching cost). The battery should be run with one pre-stated prediction per hypothesis, a common estimator, owner-clustered errors, and a correction for testing thirty comparisons at once; the paper would then report where the evidence lines up with each literature, not a search for significance. Owner locations are the applicant’s ZIP code as filed, re-parsed from the annual case-file archives and placed at Census ZIP centroids (99.8% of US owners placed).

## Where being leading costs: the factors together

*For the average registration, leading costs 1.1 points of survival once the factors are held; without them, 2.3.* About half of the raw penalty comes from who files leading language and what it describes: leading filings are more often self-filed, more often platforms, and more often use the vocabulary of general-purpose technologies, and all three survive less.

*Four factors change the cost of leading after adjustment for multiple testing.* Counsel removes it: among counsel-represented registrations leading costs nothing (0.0 points), among self-filed ones 5.0. Platform and network offerings and general-purpose-technology vocabulary raise it, to 6.1 and 2.9 points. And it has grown over time: one SD of filing year (about five years) adds 0.8 points. Figure 1 draws each of these, with the class groups whose difference from all other classes is also significant.

*Patenting marks a larger, not a more durable, portfolio.* Owners who patent survive the proof 1.7 points less often, and owners with 25 or more earlier filings 10.3 points less often: large portfolios let more marks lapse. Among patenting owners leading costs 1.0 point, the same as among others (difference +0.1).

*Booms and busts do not change the cost of leading.* Against years that were neither, leading costs 0.1 points less in boom years (1998–2000, 2005–07) and 0.2 points more in bust years (2001–02, 2008–09), neither distinguishable from zero. Without the filing-year trend both appeared to reduce the penalty, because the years that were neither are mostly 2010–19, when the penalty was largest. With the trend held, growth in a class’s filing volume also no longer passes the adjustment (−0.6 points per SD, s.e. 0.22).

*Geography moves survival, not the cost of leading.* Each SD of a theme’s co-location lowers survival by 1.6 points and each SD of growth from first-time filers by 1.3; neither changes the lead effect, and being in a hub changes neither survival nor the lead effect.

*Against all other classes, alcohol, tobacco and finance survive more and pay more for leading.* With only registration-year fixed effects, registrations in classes 33, 34 and 36 survive 4.4 points more often than those in all other classes, and leading costs them 2.9 points against 0.4 elsewhere. The composite group where leading costs most also survives more (5.7 points) while paying 5.1 points for leading; in the group where leading costs least, leading is worth 7.6 points, but Figure 1 shows the shape is an inverted U: survival rises from the most lagging decile (41%) to a peak near the 65th percentile (53%) and falls back in the most leading decile (46%). In those classes the cost is to lagging, not a reward to leading first.

*Beer and tobacco differ by what their leading filings described.* In class 32 the lead effect is +9.6 points; in class 34, −12.8. The class-34 penalty is confined to registrations of 2013–2018 (−23.4; +4.1 and +0.5 in 2002–07 and 2008–12), when e-cigarette and vaping language went from none of the class’s registrations to 46%, and its most leading theme is the e-cigarette theme. Vaping registrations survive the proof 15.8 points less often than other class-34 registrations, and within each product — vaping, cigars, cigarettes, hookah — leading is not penalized (point estimates +6 to +25). The class-level penalty is a between-product effect: leading language in class 34 meant vaping, and vaping marks died. In class 32 the rising product was beer (33% of registrations in 2002–07, 67% in 2013–18), beer registrations survive 7.5 points more often than the class’s others, and leading beer filings survive 10.1 points more often than lagging ones. The record does not say why vaping marks died; the timing coincides with federal regulation of e-cigarettes as tobacco products from 2016 and the premarket review that followed, which is a hypothesis the data here do not test.

## Where being leading costs: one model

One linear probability model of surviving the five-year proof (percentage points), with class  $\times$  registration-year fixed effects and standard errors clustered by owner; 3,104,534 registrations, 2002–2018. *Lead* is the filing’s percentile of lead within its class and year, so a lead effect is the survival difference between the most leading and the most lagging filing (negative: leading costs). Every

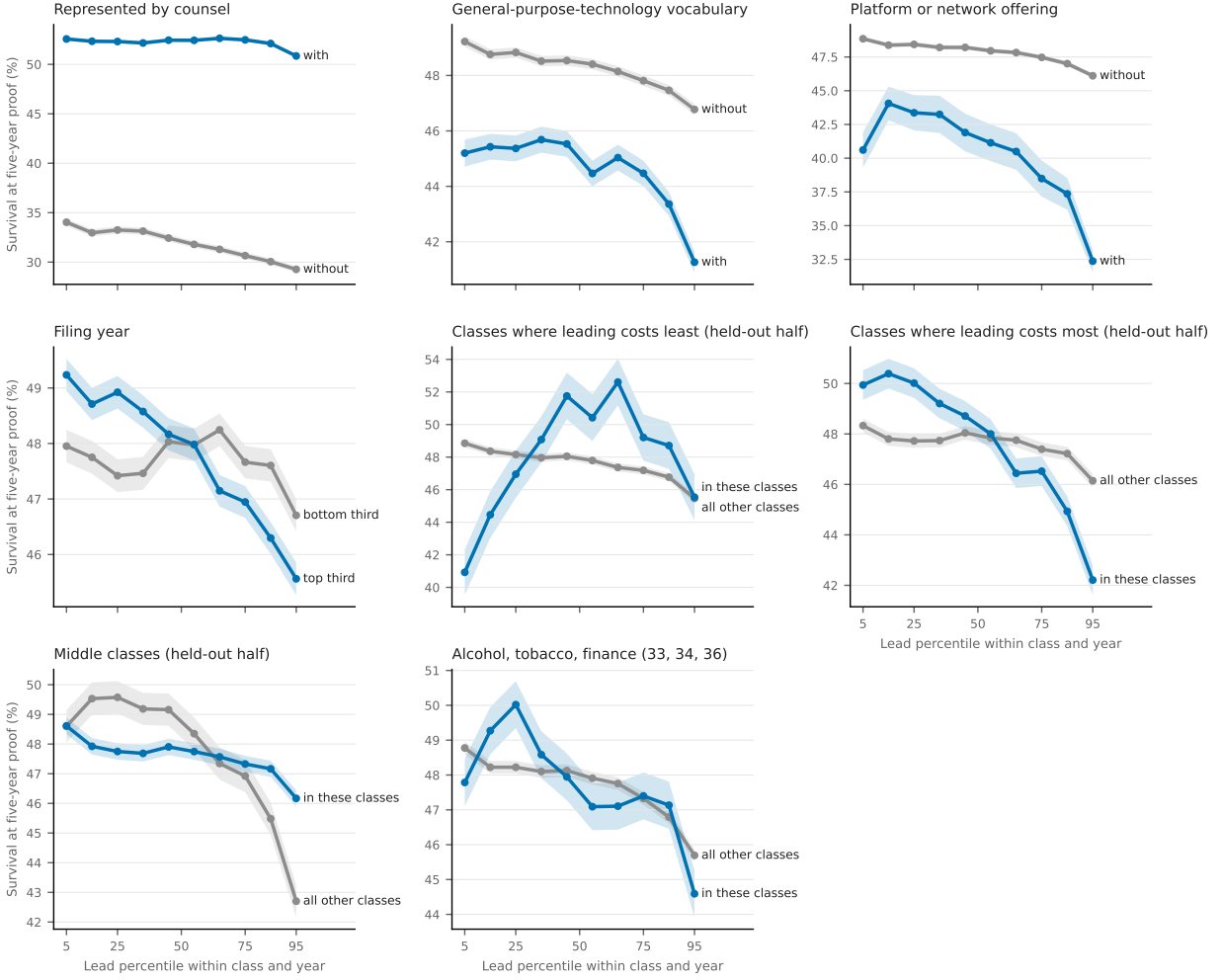


Figure 1: Survival of the five-year proof by lead decile, for registrations with and without each attribute whose with-minus-without difference in the lead effect is significant after Holm adjustment (filing year: top against bottom third; class groups: against all other classes, the composite groups on the held-out half of owners). Survival is adjusted for class  $\times$  registration-year fixed effects; bands are 95% intervals of the decile means. Vertical scales differ by panel.

factor enters as a main effect and as an interaction with lead, all at once. *Survival* is the factor's association with survival at median lead. *With* and *without* are the lead effects among filings that have and lack the factor, with standard errors (continuous factors: one SD above and below the mean; boom and bust: against years that are neither). *Difference* is with minus without; it is the test of whether the factor changes the cost of leading. Stars (Survival and Difference only): Holm-adjusted  $p$  within the column, \*  $< 0.05$ , \*\*  $< 0.01$ , \*\*\*  $< 0.001$ . The with and without estimates carry no stars because they answer a different question – is leading costly within this subgroup? – and a small subgroup can be indistinguishable from zero while the large remainder, whose estimate sits near the average of  $-1.1$ , is not. Controls: description length, owner filing count, foreign filing basis, missing-data flags.

| Factor                                                                       | Share | Survival  | Lead effect  |              |            |
|------------------------------------------------------------------------------|-------|-----------|--------------|--------------|------------|
|                                                                              |       |           | With         | Without      | Difference |
| <b>Lead, average filing</b>                                                  |       |           | -1.08 (0.24) |              |            |
| lead alone, no factors                                                       |       |           | -2.30 (0.19) |              |            |
| <i>The owner and the filing</i>                                              |       |           |              |              |            |
| Represented by counsel                                                       | 78%   | +18.76*** | +0.04 (0.27) | -5.01 (0.35) | +5.05***   |
| Owner files patents                                                          | 20%   | -1.72***  | -1.00 (0.73) | -1.10 (0.21) | +0.11      |
| Owner's first-ever filing                                                    | 33%   | -1.12     | -0.64 (0.32) | -1.30 (0.25) | +0.65      |
| Owner has 25+ earlier filings                                                | 20%   | -10.26*** | -1.20 (0.71) | -1.05 (0.21) | -0.15      |
| Foreign owner (not China)                                                    | 18%   | +0.99     | -0.40 (0.55) | -1.23 (0.28) | +0.84      |
| Owner in China                                                               | 3%    | +1.04     | +0.50 (0.88) | -1.14 (0.25) | +1.63      |
| Intent-to-use filing                                                         | 41%   | -6.81***  | -1.61 (0.37) | -0.71 (0.23) | -0.90      |
| <i>The offering</i>                                                          |       |           |              |              |            |
| Site-based offering (stores, restaurants, clinics)                           | 9%    | +2.90***  | -2.15 (0.49) | -0.98 (0.25) | -1.17      |
| Contract or subscription offering                                            | 5%    | -0.48     | +0.90 (0.81) | -1.19 (0.25) | +2.10      |
| Platform or network offering                                                 | 2%    | -5.70***  | -6.10 (1.12) | -0.99 (0.25) | -5.11***   |
| Invention vocabulary                                                         | 0.4%  | -6.59***  | -1.92 (1.93) | -1.08 (0.24) | -0.84      |
| General-purpose-technology vocabulary                                        | 15%   | -5.13***  | -2.88 (0.41) | -0.77 (0.26) | -2.11***   |
| Emergent consumer-category vocabulary                                        | 0.4%  | -12.56*** | -1.05 (1.84) | -1.08 (0.25) | +0.03      |
| Theme imported from another class                                            | 5%    | +1.98***  | -1.35 (0.62) | -1.07 (0.24) | -0.29      |
| <i>The theme and its geography</i>                                           |       |           |              |              |            |
| Theme-share volatility (per SD)                                              | SD    | +0.09     | -0.76 (0.23) | -1.40 (0.41) | +0.64      |
| Theme growth from first-time filers (per SD)                                 | SD    | -1.32***  | -1.16 (0.26) | -1.00 (0.43) | -0.16      |
| Theme co-location L (per SD)                                                 | SD    | -1.57***  | -1.36 (0.32) | -0.80 (0.30) | -0.56      |
| Theme has a geographic hub                                                   | 28%   | -0.12     | -1.10 (0.34) | -1.07 (0.30) | -0.03      |
| Filer located in a hub                                                       | 5%    | +0.26     | -1.03 (0.82) | -1.08 (0.23) | +0.05      |
| <i>The class and the year (boom and bust against years that are neither)</i> |       |           |              |              |            |
| Class theme-mix turnover (per SD)                                            | SD    | +0.46**   | -1.51 (0.38) | -0.65 (0.22) | -0.87      |
| Class filing-volume growth (per SD)                                          | SD    | +0.24     | -1.64 (0.25) | -0.52 (0.39) | -1.12      |
| Filing year (per SD, about 5 years)                                          | SD    | +0.72     | -1.87 (0.42) | -0.29 (0.27) | -1.58*     |

| Factor                               | Share | Survival | Lead effect  |              |            |
|--------------------------------------|-------|----------|--------------|--------------|------------|
|                                      |       |          | With         | Without      | Difference |
| Filed in a boom (1998-2000, 2005-07) | 19%   | -0.37    | -0.95 (0.48) | -1.06 (0.25) | +0.11      |
| Filed in a bust (2001-02, 2008-09)   | 18%   | -0.04    | -1.29 (0.40) | -1.06 (0.25) | -0.22      |

**Class groups against all other classes.** Class fixed effects absorb a class group’s survival level, so the groups are compared here with all other classes in a model with registration-year fixed effects only, the factors above and their lead interactions held. Survival is the group’s difference from all other classes; the lead effects are within the group and within all other classes. The three composite groups (next section) are estimated on the half of owners that played no part in forming them.

| Class group                                       | Share | Survival | Lead effect  |               |            |
|---------------------------------------------------|-------|----------|--------------|---------------|------------|
|                                                   |       |          | In group     | Other classes | Difference |
| Services classes (35-45)                          | 37%   | -1.21*** | -0.29 (0.32) | -0.62 (0.37)  | +0.33      |
| Pharmaceuticals and devices (5, 10)               | 5%    | -3.90*** | -1.50 (1.48) | -0.21 (0.27)  | -1.29      |
| Consumer packaged goods (3, 29-33)                | 12%   | -0.07    | -1.39 (0.61) | -0.24 (0.27)  | -1.15      |
| Machinery, electronics, vehicles (7, 9, 12)       | 17%   | +1.84*** | -0.81 (0.43) | -0.31 (0.29)  | -0.49      |
| Alcohol, tobacco, finance (33, 34, 36)            | 7%    | +4.41*** | -2.86 (0.71) | -0.36 (0.27)  | -2.49*     |
| Classes where leading costs most (held-out half)  | 17%   | +5.71*** | -5.13 (0.53) | +0.37 (0.47)  | -5.51***   |
| Middle classes (held-out half)                    | 80%   | -5.21*** | +0.08 (0.48) | -3.21 (0.47)  | +3.29***   |
| Classes where leading costs least (held-out half) | 3%    | +0.39    | +7.60 (1.18) | -0.53 (0.39)  | +8.12***   |

**Each factor alone.** The survival and lead-interaction coefficients from models with lead, one factor and the controls (boom and bust together with the trend), against the joint model (Holm stars within column).

| Factor                                             | Survival  |           | Lead × factor |          |
|----------------------------------------------------|-----------|-----------|---------------|----------|
|                                                    | Alone     | Joint     | Alone         | Joint    |
| Represented by counsel                             | +18.80*** | +18.76*** | +4.85***      | +5.05*** |
| Owner files patents                                | -2.74***  | -1.72***  | +0.88         | +0.11    |
| Owner’s first-ever filing                          | -5.14***  | -1.12     | -0.25         | +0.65    |
| Owner has 25+ earlier filings                      | -12.38*** | -10.26*** | +0.77         | -0.15    |
| Foreign owner (not China)                          | +2.93***  | +0.99     | +1.28         | +0.84    |
| Owner in China                                     | -3.42***  | +1.04     | -0.27         | +1.63    |
| Intent-to-use filing                               | -5.99***  | -6.81***  | +0.23         | -0.90    |
| Site-based offering (stores, restaurants, clinics) | +2.74***  | +2.90***  | -1.57*        | -1.17    |
| Contract or subscription offering                  | -0.85     | -0.48     | +2.33         | +2.10    |
| Platform or network offering                       | -9.18***  | -5.70***  | -8.41***      | -5.11*** |
| Invention vocabulary                               | -8.18***  | -6.59***  | -2.45         | -0.84    |

| Factor                                       | Survival  |           | Lead $\times$ factor |          |
|----------------------------------------------|-----------|-----------|----------------------|----------|
|                                              | Alone     | Joint     | Alone                | Joint    |
| General-purpose-technology vocabulary        | -5.91***  | -5.13***  | -1.92***             | -2.11*** |
| Emergent consumer-category vocabulary        | -13.82*** | -12.56*** | +0.70                | +0.03    |
| Theme imported from another class            | +3.17***  | +1.98***  | -0.37                | -0.29    |
| Theme-share volatility (per SD)              | +0.50***  | +0.09     | -0.43                | +0.32    |
| Theme growth from first-time filers (per SD) | -1.90***  | -1.32***  | -0.12                | -0.08    |
| Theme co-location L (per SD)                 | -2.03***  | -1.57***  | -0.43                | -0.28    |
| Theme has a geographic hub                   | -1.71***  | -0.12     | -0.84                | -0.03    |
| Filer located in a hub                       | -1.93***  | +0.26     | -0.47                | +0.05    |
| Class theme-mix turnover (per SD)            | +0.64***  | +0.46**   | -1.12***             | -0.43    |
| Class filing-volume growth (per SD)          | +0.15     | +0.24     | -1.59***             | -0.56    |
| Filing year (per SD, about 5 years)          | +3.73***  | +0.72     | -1.81***             | -0.79*   |
| Filed in a boom (1998-2000, 2005-07)         | -0.50*    | -0.37     | -0.72                | +0.11    |
| Filed in a bust (2001-02, 2008-09)           | -0.30     | -0.04     | -0.23                | -0.22    |

## Classes that behave alike

Class-specific lead effects were estimated on a random half of owners (split by owner name), shrunk toward the pooled effect in proportion to their noise, and grouped into three by  $k$ -means. The groups' lead effects below are estimated on the other half of owners, which played no part in forming them.

| Group                             | Nice classes (shrunk lead effect, first half)                                                                                                                                                                                                                                                                             | Lead effect, second half |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Classes where leading costs most  | 34 (-11.4), 23 (-7.9), 33 (-7.8), 42 (-7.6), 10 (-7.5), 4 (-6.0), 11 (-5.9), 29 (-5.7), 27 (-5.0), 1 (-4.8), 6 (-4.0), 7 (-3.5)                                                                                                                                                                                           | -5.51*** (s.e. 0.50)     |
| Middle classes                    | 17 (-3.1), 9 (-3.0), 43 (-3.0), 13 (-2.5), 20 (-2.2), 16 (-1.9), 12 (-1.6), 24 (-1.6), 22 (-1.4), 5 (-1.0), 31 (-0.8), 8 (-0.6), 38 (-0.5), 28 (-0.4), 30 (-0.4), 19 (-0.4), 35 (-0.2), 41 (-0.1), 40 (+0.1), 14 (+0.4), 3 (+0.7), 21 (+1.0), 36 (+1.1), 15 (+1.2), 25 (+1.9), 39 (+2.5), 37 (+2.5), 44 (+2.8), 18 (+3.3) | -0.35 (s.e. 0.42)        |
| Classes where leading costs least | 2 (+5.6), 45 (+7.3), 26 (+8.8), 32 (+10.3)                                                                                                                                                                                                                                                                                | +7.09*** (s.e. 1.11)     |

## Which factors explain where leading costs (LASSO)

With fixed effects, main effects and controls partialled out, a LASSO was fitted over every lead interaction: the factors above, the 45 classes and the three class groups (all standardized), on 1,250,681 registrations from a random 40% of owners. The penalty was chosen by five-fold cross-validation with folds grouped by owner. Terms are listed in the order they enter as the penalty is relaxed, the usual ranking of explanatory value. At the cross-validated penalty 39 of the 77

candidate terms are kept (61 enter anywhere on the path); the stricter one-standard-error rule keeps none, because for a pass/fail outcome the gain in out-of-sample fit is within the fold-to-fold noise. The last column re-estimates the first 10 terms to enter, jointly, on all registrations with clustered errors (after selection, so the stars are optimistic).

| Order | Lead interaction                      | Re-estimated (pp) |
|-------|---------------------------------------|-------------------|
| 1     | Represented by counsel                | +5.01*** (0.37)   |
| 2     | Nice class 28                         | +6.82 (4.99)      |
| 3     | Classes where leading costs most      | -4.12*** (0.45)   |
| 4     | Class filing-volume growth (per SD)   | -0.96*** (0.17)   |
| 5     | Platform or network offering          | -5.57*** (1.12)   |
| 6     | Classes where leading costs least     | +8.22*** (0.94)   |
| 7     | Nice class 34                         | -8.01* (3.05)     |
| 8     | General-purpose-technology vocabulary | -1.79*** (0.44)   |
| 9     | Nice class 42                         | -2.50** (0.74)    |
| 10    | Nice class 32                         | +2.33 (1.58)      |
| 11    | Nice class 33                         |                   |
| 12    | Nice class 39                         |                   |
| 13    | Nice class 43                         |                   |
| 14    | Theme co-location L (per SD)          |                   |
| 15    | Nice class 38                         |                   |
| 16    | Nice class 6                          |                   |
| 17    | Nice class 37                         |                   |
| 18    | Nice class 10                         |                   |
| 19    | Nice class 31                         |                   |
| 20    | Filed in a boom (1998-2000, 2005-07)  |                   |

## Classes that look alike

Each Nice class is described by five metrics computed on its 2002–2018 registrations: the share passing the five-year proof; the SD of that share across registration years; the mean and variance of the raw lead score; and froth, the class’s average five-year turnover in its theme mix (half the summed absolute change in theme shares between the five filing years before and the five after; 0 = no change, 1 = complete). The metrics were standardized and clustered by Ward linkage;  $k = 4$  has the highest silhouette (0.26; values near 0.25 indicate weak separation, so the clusters are a suggestion for review, not a finding). The last column, each class’s own lead effect (survival of its most leading minus its most lagging filing, class  $\times$  year fixed effects and the paper’s controls), played no part in forming the clusters.

The clusters account for 26% of the precision-weighted variation in class lead effects ( $\chi^2_3 = 148$ ,  $p < 10^{-31}$ ). Across classes, the lead effect correlates (precision-weighted) with survival +0.12; sd across years -0.15; lead, mean -0.08; lead, variance -0.54; froth -0.40.

| Class                                                                                        | Survival | SD yrs | Lead mean | Lead var. | Froth | Lead effect |
|----------------------------------------------------------------------------------------------|----------|--------|-----------|-----------|-------|-------------|
| <b>Cluster 1: widely spread lead, frothy</b> (5 classes; pooled lead effect -6.9, s.e. 0.48) |          |        |           |           |       |             |
| 42 Technology, science services                                                              | 43.5     | 2.4    | 0.012     | 0.031     | 0.094 | -10.4 (0.6) |
| 26 Lace, ribbons                                                                             | 43.5     | 4.5    | 0.062     | 0.024     | 0.112 | -0.6 (2.5)  |
| 21 Housewares                                                                                | 45.9     | 2.6    | 0.036     | 0.031     | 0.100 | -1.3 (1.1)  |
| 40 Treatment of materials                                                                    | 49.6     | 2.3    | -0.001    | 0.018     | 0.088 | +1.5 (1.4)  |

| Class                                                                                      | Survival | SD yrs | Lead mean | Lead var. | Froth | Lead effect |
|--------------------------------------------------------------------------------------------|----------|--------|-----------|-----------|-------|-------------|
| 4 Lubricants, fuels                                                                        | 50.4     | 3.1    | −0.010    | 0.028     | 0.092 | −3.1 (2.6)  |
| <b>Cluster 2: volatile survival, frothy</b> (1 class; pooled lead effect −12.8, s.e. 2.89) |          |        |           |           |       |             |
| 34 Tobacco, vaping                                                                         | 45.9     | 8.1    | −0.002    | 0.022     | 0.140 | −12.8 (2.9) |
| <b>Cluster 3: high survival, calm</b> (11 classes; pooled lead effect −0.5, s.e. 0.40)     |          |        |           |           |       |             |
| 12 Vehicles                                                                                | 54.1     | 2.0    | 0.006     | 0.003     | 0.041 | 0.4 (1.2)   |
| 19 Building materials                                                                      | 56.0     | 2.1    | −0.000    | 0.002     | 0.035 | 0.5 (1.5)   |
| 31 Agricultural produce                                                                    | 57.2     | 2.0    | −0.007    | 0.003     | 0.043 | −4.6 (1.8)  |
| 33 Wine, spirits                                                                           | 58.6     | 1.6    | 0.000     | 0.002     | 0.039 | −0.8 (1.2)  |
| 15 Musical instruments                                                                     | 58.9     | 4.3    | 0.008     | 0.002     | 0.043 | 0.7 (2.5)   |
| 2 Paints                                                                                   | 59.4     | 3.3    | 0.002     | 0.010     | 0.052 | +4.9 (2.0)  |
| 7 Machinery                                                                                | 60.7     | 1.5    | 0.002     | 0.007     | 0.060 | −1.9 (0.9)  |
| 1 Chemicals                                                                                | 60.8     | 1.1    | −0.001    | 0.007     | 0.057 | −1.5 (1.0)  |
| 6 Metal goods                                                                              | 60.8     | 1.6    | 0.004     | 0.004     | 0.048 | 0.0 (1.0)   |
| 17 Rubber, plastics                                                                        | 61.6     | 2.3    | 0.003     | 0.002     | 0.037 | 0.3 (1.6)   |
| 13 Firearms                                                                                | 62.9     | 4.1    | 0.007     | 0.006     | 0.066 | +3.0 (2.6)  |
| <b>Cluster 4: low survival</b> (28 classes; pooled lead effect −1.2, s.e. 0.16)            |          |        |           |           |       |             |
| 25 Clothing                                                                                | 39.7     | 2.6    | 0.007     | 0.001     | 0.021 | −2.5 (0.5)  |
| 38 Telecommunications                                                                      | 39.8     | 3.6    | −0.006    | 0.010     | 0.069 | −0.3 (1.3)  |
| 28 Toys, sporting goods                                                                    | 42.4     | 3.5    | 0.005     | 0.008     | 0.059 | +4.0 (2.0)  |
| 18 Leather goods                                                                           | 42.5     | 2.6    | 0.019     | 0.003     | 0.043 | −0.0 (1.0)  |
| 14 Jewelry                                                                                 | 43.3     | 3.3    | 0.011     | 0.006     | 0.061 | −2.0 (1.0)  |
| 45 Legal, personal services                                                                | 44.3     | 2.6    | 0.019     | 0.008     | 0.068 | +2.5 (1.1)  |
| 23 Yarns, threads                                                                          | 44.7     | 5.8    | −0.001    | 0.008     | 0.058 | −8.2 (4.9)  |
| 41 Education, entertainment                                                                | 44.7     | 1.7    | 0.008     | 0.004     | 0.046 | −0.0 (0.5)  |
| 3 Cosmetics, cleaning                                                                      | 45.0     | 1.7    | −0.001    | 0.004     | 0.037 | −3.8 (0.9)  |
| 35 Advertising, business                                                                   | 45.0     | 2.2    | −0.002    | 0.005     | 0.050 | −0.4 (0.4)  |
| 24 Fabrics                                                                                 | 45.9     | 2.3    | 0.016     | 0.007     | 0.060 | −5.8 (1.7)  |
| 5 Pharmaceuticals                                                                          | 45.9     | 1.5    | −0.001    | 0.007     | 0.055 | −4.0 (1.4)  |
| 44 Medical, beauty services                                                                | 46.7     | 2.3    | 0.005     | 0.003     | 0.055 | −0.5 (0.9)  |
| 32 Beer, soft drinks                                                                       | 46.9     | 3.0    | 0.001     | 0.003     | 0.048 | +9.6 (1.3)  |
| 9 Electronics, software                                                                    | 47.1     | 2.4    | 0.002     | 0.012     | 0.072 | −3.1 (0.5)  |
| 30 Coffee, bakery, staples                                                                 | 47.5     | 2.0    | −0.001    | 0.003     | 0.033 | −1.7 (0.9)  |
| 16 Paper, printed matter                                                                   | 48.3     | 1.1    | 0.007     | 0.007     | 0.056 | 0.3 (0.7)   |
| 20 Furniture                                                                               | 48.8     | 2.0    | 0.018     | 0.009     | 0.069 | −2.3 (1.0)  |
| 39 Transport, storage                                                                      | 50.1     | 3.2    | −0.000    | 0.004     | 0.051 | +4.6 (1.3)  |
| 36 Insurance, finance                                                                      | 50.2     | 2.2    | 0.001     | 0.009     | 0.055 | 0.7 (0.7)   |
| 29 Meat, dairy, processed                                                                  | 50.7     | 2.6    | −0.003    | 0.002     | 0.043 | −6.6 (1.2)  |
| 43 Hotels, restaurants                                                                     | 50.8     | 4.6    | −0.005    | 0.002     | 0.057 | −4.1 (0.9)  |
| 37 Construction, repair                                                                    | 51.0     | 1.8    | 0.013     | 0.009     | 0.057 | +4.0 (1.0)  |
| 27 Floor coverings                                                                         | 51.6     | 4.3    | −0.007    | 0.011     | 0.076 | −1.6 (3.3)  |
| 22 Rope, fibers                                                                            | 53.3     | 4.2    | 0.010     | 0.006     | 0.066 | −3.2 (2.8)  |
| 11 Lighting, heating                                                                       | 53.3     | 2.4    | 0.015     | 0.011     | 0.077 | −5.5 (1.1)  |
| 8 Hand tools                                                                               | 54.3     | 2.8    | 0.009     | 0.008     | 0.064 | −5.2 (1.6)  |
| 10 Medical devices                                                                         | 55.7     | 2.3    | 0.004     | 0.014     | 0.068 | −6.9 (1.0)  |

## What the battery shows

Across all registrations of 2002–2018 the most leading fifth fails the five-year proof 1.39 points more often than the most lagging fifth. Six of the thirty pre-stated contrasts survive adjustment for multiple testing; four of the six run against the prediction.

*The penalty is concentrated where a class’s filing volume is growing.* Where both the class’s theme mix and its filing volume changed fast, the penalty is 3.2 points; where only volume grew, 1.9; in calm class-years, 0.6; where only the mix turned over,  $-0.2$ . The pace-of-change account predicts the most durable advantage where the market grows and the technology is stable (Suárez & Lanzolla, 2007); the record puts the penalty there instead. This is what the crowding reading of lead predicts: when entrants pour into a class, the language they pour into is the language that fails.

*Platform and network offerings carry the largest penalty.* Filings describing marketplaces, platforms, social networks or peer-to-peer services have a penalty of 7.2 points, against 1.2 for all other filings. Network effects are listed as a source of first-mover advantage (Lieberman & Montgomery, 1988); in the record, early entrants to networked categories fail more often than late ones.

*In consumer packaged goods the penalty falls on established firms, not new ones.* Among owners with 25 or more earlier filings, leading filings fail 4.9 points more often; among first-time filers, 0.9 points less often. The complementary-assets account predicts the reverse (Teece, 1986): incumbents with distribution should hold what small pioneers open.

*Manufactured goods carry a larger penalty than services* (2.3 against 1.1 points), the reverse of the shakeout account, in which early manufacturers are the likeliest survivors (Klepper, 1997; Klepper & Miller, 1995).

*Counsel absorbs most of the penalty.* Self-filed leading filings fail 4.2 points more often than self-filed lagging ones; counsel-represented, 0.6.

*Two contrasts fall just short of the adjustment* (Holm  $p = 0.065$ ): the penalty is larger in themes whose share swings from year to year (2.1 against 0.9), as the uncertainty account predicts, and larger for site-based offerings such as restaurants, hotels and clinics (2.7 against 1.4), against the preemption account.

*Pioneers of a new vocabulary fail, and so do its early followers.* Filings using a new technology’s or product’s vocabulary in its first two years in a class fail 12.8 points more often than other filings of the same class and year; filings two to five years in, 11.8 points more often; filings six or more years in, 6.0. The first two do not differ ( $t = 0.5$ ). Golder and Tellis’s pioneers failed; here the early followers failed at the same rate.

*Within verticals the sign differs.* Leading filings in beer fail 6.2 points less often than lagging ones, and in live entertainment 5.6 points less often; in coffee they fail 3.2 points more often and in fitness 2.9. These vertical estimates are not adjusted for multiple testing.

*The blue-ocean exemplars’ own marks do not lead their classes.* The mean lead fifth of the registrations of [yellow tail]’s owner, Cirque du Soleil, Southwest Airlines, NetJets, Accor, Starbucks and Home Depot lies between 2.7 and 3.2, the middle of the distribution; only Curves’s marks lean leading (3.8). These are the firms’ marks of 2002–2018, mostly filed years after each created its category.

*Geography changes where filers fail, not the lead penalty.* A theme’s co-location with itself does not change its penalty (1.8 in the most concentrated third, 2.3 in the most dispersed). Within themes that have a hub, filers in the hub fail 1.2 points more often than filers outside it (s.e. 0.45), whatever their lead, and their lead penalty is the same.

*Nothing distinguishes patenting owners, pharmaceuticals and devices, consumer against industrial goods, themes imported from another class, themes grown by first-time filers, invention*

vocabularies, fads against surges that held, vocabulary new to its class against new to the corpus, foreign owners, intent-to-use filings, regulated classes, or boom against bust years. Several of these groups are small (invention vocabularies 12,612 registrations; general-purpose-technology filings in their first three years in a class, 1,166).

## Results

Penalty = failure rate at the five-year proof of the most leading fifth minus the most lagging fifth, in percentage points (all registrations 2002–2018: +1.39, s.e. 0.17). Each row’s two penalties come from one model with the lead fifths interacted with the group; the difference is the first minus the second. Predicted: the sign of that difference stated in the catalogue (? = no prediction). Holm  $p$  adjusts across all pre-stated contrasts. Agrees: whether the sign matches the prediction, given only where Holm  $p < 0.10$ .

|    | Groups compared                                                                                  | Penalty (pp)                               | Diff. | $t$  | Holm $p$ | Pred. | Agrees |
|----|--------------------------------------------------------------------------------------------------|--------------------------------------------|-------|------|----------|-------|--------|
| 1  | Owners who patent vs those who do not                                                            | patenter +1.10; no patents +1.50           | -0.39 | -0.6 | 1.000    | -     | –      |
| 2  | Site-based offerings vs others                                                                   | site-based +2.68; other +1.40              | +1.28 | 3.0  | 0.065    | -     | no     |
| 3  | Contract / subscription offerings vs others                                                      | switching costs -0.32; other +1.51         | -1.83 | -2.7 | 0.159    | -     | –      |
| 4  | Platform / network offerings vs others                                                           | network +7.17; other +1.17                 | +6.00 | 5.8  | 0.000    | -     | no     |
| 5  | Easy-to-imitate goods vs capital-heavy goods                                                     | easy to imitate -0.01; capital heavy +1.53 | -1.54 | -2.5 | 0.232    | +     | –      |
| 6  | Volatile themes vs steady themes (top vs bottom third of share volatility)                       | volatile +2.10; steady +0.93               | +1.18 | 3.0  | 0.065    | +     | yes    |
| 7  | Debut owners vs established owners (25+ prior filings)                                           | debut +1.73; established +1.36             | +0.37 | 0.6  | 1.000    | +     | –      |
| 8  | Pace: technology leads vs calm                                                                   | technology leads -0.24; calm +0.55         | -0.78 | -2.2 | 0.595    | +     | –      |
| 10 | Pace: market leads vs technology leads                                                           | market leads +1.92; technology leads -0.24 | +2.15 | 5.4  | 0.000    | -     | no     |
| 11 | Pace: rough vs calm                                                                              | rough +3.21; calm +0.55                    | +2.66 | 7.0  | 0.000    | +     | yes    |
| 12 | Pharmaceuticals and medical devices (classes 5, 10) vs all other classes                         | pharma/devices +2.88; other +1.31          | +1.57 | 1.3  | 1.000    | -     | –      |
| 13 | Services (classes 35-45) vs goods                                                                | services +1.12; goods +1.56                | -0.43 | -1.3 | 1.000    | +     | –      |
| 14 | Consumer packaged goods: debut owners vs established owners                                      | debut -0.86; established +4.89             | -5.75 | -4.6 | 0.000    | +     | no     |
| 15 | Pioneers of a new vocabulary in a class vs early followers (excess failure over same class-year) | pioneer –; early follower –                | +0.95 | 0.5  | 1.000    | +     | –      |

|    | Groups compared                                                                              | Penalty (pp)                                   | Diff. | $t$  | Holm $p$ | Pred. | Agrees |
|----|----------------------------------------------------------------------------------------------|------------------------------------------------|-------|------|----------|-------|--------|
| 17 | Consumer packaged goods vs industrial goods                                                  | CPG +1.57; industrial +0.96                    | +0.61 | 0.9  | 1.000    | -     | -      |
| 18 | Manufactured goods (7, 9, 12) vs services (35-45)                                            | manufactured +2.33; services +1.08             | +1.25 | 3.2  | 0.034    | -     | no     |
| 20 | Theme imported from another class vs native theme                                            | imported +1.92; native +1.35                   | +0.56 | 1.3  | 1.000    | -     | -      |
| 22 | Themes growing through first-time filers vs through established filers (top vs bottom third) | new demand +1.05; established demand +0.33     | +0.72 | 1.6  | 1.000    | -     | -      |
| 23 | Invention vocabularies vs all other filings                                                  | invention +3.33; other +1.38                   | +1.95 | 1.1  | 1.000    | -     | -      |
| 24 | Fads vs surges that held                                                                     | fad (surge given back) +1.41; surge held +2.87 | -1.46 | -0.5 | 1.000    | +     | -      |
| 25 | General-purpose technology filings: first 3 years in a class vs 6+ years                     | early adopter -2.30; late adopter +1.86        | -4.16 | -0.8 | 1.000    | -     | -      |
| 26 | Vocabulary new to its class vs new to the corpus (excess failure, first five years)          | new to class +11.09; new to corpus +9.46       | +1.63 | 0.6  | 1.000    | -     | -      |
| 27 | Emergent consumer categories: first three years vs later (excess failure)                    | first 3 years +12.62; later +12.63             | -0.01 | -0.0 | 1.000    | ?     |        |
| 28 | Geographically concentrated themes vs dispersed (top vs bottom third of co-location L)       | concentrated +1.80; dispersed +2.31            | -0.51 | -1.1 | 1.000    | +     | -      |
| 29 | Within themes that have a hub: filers in a hub vs filers outside                             | in hub +2.81; outside hub +3.08                | -0.27 | -0.4 | 1.000    | ?     |        |
| 30 | Foreign-domiciled owners vs US owners                                                        | foreign (not China) +0.63; US +1.65            | -1.02 | -1.9 | 0.987    | +     | -      |
| 31 | Self-filed vs counsel-represented                                                            | self-filed +4.20; counsel +0.59                | +3.61 | 7.7  | 0.000    | +     | yes    |
| 32 | Intent-to-use vs use-based filings                                                           | intent to use +1.43; in use +1.36              | +0.07 | 0.2  | 1.000    | +     | -      |
| 33 | Regulated classes (5, 10, 33, 34, 36) vs others                                              | regulated +1.96; unregulated +1.31             | +0.65 | 1.0  | 1.000    | -     | -      |
| 34 | Filed in a boom (2004-07, 2014-17) vs a bust (2001-02, 2008-10)                              | boom +0.92; bust +1.01                         | -0.09 | -0.3 | 1.000    | +     | -      |

Rows 15, 26 and 27 report excess failure (pp) over other filings in the same class and registration year, not a lead penalty.

**Pace of change (hypotheses 8–11).** Penalty by quadrant: calm +0.55 ( $n = 830,672$ ); technology leads -0.24 ( $n = 718,425$ ); market leads +1.92 ( $n = 707,755$ ); rough +3.21 ( $n = 847,682$ ).

**Golder–Tellis cases (hypothesis 19).** Penalty within each: diapers +0.36 (s.e. 1.97,  $n = 9,788$ ); video recorders +3.65 (s.e. 3.47,  $n = 10,306$ ); personal computers +2.19 (s.e. 1.54,  $n = 27,690$ ); beer -6.16 (s.e. 1.33,  $n = 39,290$ ); razors +0.34 (s.e. 2.93,  $n = 4,090$ ).

**Blue-ocean exemplar verticals (hypothesis 21).** Penalty within each: wine +1.29 (s.e. 0.94,  $n = 49,428$ ); fitness +2.93 (s.e. 1.06,  $n = 47,894$ ); live entertainment -5.58 (s.e. 1.50,  $n = 26,045$ ); air travel +1.21 (s.e. 2.94,  $n = 5,729$ ); hotels -1.67 (s.e. 1.47,  $n = 24,879$ ); coffee +3.16 (s.e. 1.07,  $n = 38,624$ ); home improvement +1.10 (s.e. 1.96,  $n = 13,176$ ).

**The exemplar firms’ own marks.** Registrations 2002–2018 by owner; mean lead fifth (1 = most lagging, 5 = most leading), share in the most leading fifth, and proof failure rate: [yellow tail] (Casella) 109 marks, fifth 3.0, 23% leading, 48% failed; Curves 65 marks, fifth 3.8, 42% leading, 51% failed; Cirque du Soleil 64 marks, fifth 2.7, 22% leading, 34% failed; Southwest Airlines 103 marks, fifth 2.7, 12% leading, 42% failed; NetJets 9 marks, fifth 2.7, 11% leading, 100% failed; Accor (Formule 1) 284 marks, fifth 2.9, 19% leading, 45% failed; Starbucks 239 marks, fifth 3.2, 30% leading, 42% failed; Home Depot 346 marks, fifth 2.9, 18% leading, 17% failed.

**Being in a hub.** Within themes that have a hub, filers in the hub fail +1.19 points more often than filers outside it (s.e. 0.45), whatever their lead.

## The clustering measure

For a group of  $N$  US filers placed at their ZIP centroids, the co-location index is the average over pairs of distinct filers of  $k(d) = \max(0, 1 - d/200 \text{ miles})$ : two filers in the same place score 1, two 100 miles apart 0.5, two more than 200 miles apart 0. It is the chance that two filers drawn at random are neighbours, weighted by how near, and it allows any number of hubs. Filers crowd into populous places whatever they sell, so the index is set against the same index  $B$  for all filers in the class and years:  $L = (C - B)/(1 - B)$ , which is 0 for a group placed like its class and 1 when every filer is in one place. This is the distance-based comparison of Duranton & Overman (2005) summarized at one bandwidth. A hub is a place where at least 10% of the group’s filers are nearby (kernel-weighted) and at least 1.2 times the class’s share; hubs are picked greedily and named by the commonest owner city within 50 miles. For the battery the groups are each class’s filers of one primary theme in a five-year window around the filing year, with at least 30 US filers.

| Group                                              | $N$    | $C$   | $B$   | $L$   | Hubs (share nearby vs class)                                                     |
|----------------------------------------------------|--------|-------|-------|-------|----------------------------------------------------------------------------------|
| internet vocabulary, classes 9/35/38/42, 1998-2001 | 60,903 | 0.075 | 0.062 | 0.014 | San Francisco, CA 13% vs 10%                                                     |
| semiconductors, class 9, 1995-2005                 | 4,517  | 0.254 | 0.064 | 0.203 | Santa Clara, CA 48% vs 12%                                                       |
| wine, class 33, 2000-2010                          | 18,799 | 0.159 | 0.108 | 0.056 | Napa, CA 35% vs 26%                                                              |
| country music, class 41, 1995-2010                 | 163    | 0.106 | 0.067 | 0.041 | Nashville, TN 28% vs 1%                                                          |
| casino gaming, class 41, 1995-2010                 | 4,662  | 0.156 | 0.067 | 0.095 | Las Vegas, NV 37% vs 2%                                                          |
| biotechnology, classes 1/5/42, 1995-2010           | 2,695  | 0.090 | 0.060 | 0.031 | Philadelphia, PA 16% vs 13%; Foster City, CA 14% vs 7%; New Haven, CT 13% vs 11% |

| Group                                          | <i>N</i> | <i>C</i> | <i>B</i> | <i>L</i> | Hubs (share nearby vs class)                                                    |
|------------------------------------------------|----------|----------|----------|----------|---------------------------------------------------------------------------------|
| social networking, classes 38/42/45, 2004-2012 | 14,041   | 0.088    | 0.057    | 0.033    | New York, NY 18% vs 14%; San Francisco, CA 15% vs 8%; Los Angeles, CA 13% vs 9% |
| online advertising, classes 35/42, 2005-2015   | 12,028   | 0.079    | 0.054    | 0.027    | New York, NY 17% vs 14%; San Francisco, CA 14% vs 7%; Los Angeles, CA 12% vs 9% |
| software as a service, class 42, 2008-2016     | 20,783   | 0.064    | 0.057    | 0.008    | San Francisco, CA 15% vs 10%                                                    |
| hip hop, classes 9/25/41, 1995-2010            | 507      | 0.122    | 0.068    | 0.058    | New York, NY 29% vs 16%                                                         |

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